

Machine Learning-Based Carbon Intensity Forecasting and Carbon-Aware Task Scheduling for Sustainable Cloud Computing in Bangladesh

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Abstract

Carbon-aware computing — shifting delay-tolerant workloads to periods when the electricity grid is powered by cleaner sources — has emerged as a major strategy for reducing the carbon footprint of cloud computing. Existing frameworks (Google's Carbon-Intelligent Compute Management, Microsoft's Carbon-Aware SDK) rely on mature, fine-grained regional carbon-intensity signals that are largely unavailable for South Asian grids, including Bangladesh's. This paper (1) constructs an hourly carbon-intensity index for Bangladesh's grid from the real PGCB (Power Grid Company of Bangladesh) hourly generation dataset, (2) trains and evaluates a Random Forest forecasting model against classical baselines, and (3) designs, implements, and simulates a load-aware carbon-aware task scheduler, comparing it against three baseline scheduling policies on a shared job stream. Using 21,565 real hourly records (January 2023 to June 2025), results show that Linear Regression achieves the best forecasting accuracy (MAE 8.28 gCO₂/kWh), narrowly ahead of a Random Forest model (MAE 9.32) and well ahead of a naïve persistence baseline (MAE 13.93). The proposed carbon-aware scheduler reduces total carbon emissions by 0.55% relative to a run-immediately baseline — a modest but genuine and reproducible figure, consistent with Bangladesh's still-limited (though measurably rising) grid-carbon-intensity variability. This is, to our knowledge, the first study to apply carbon-aware forecasting and scheduling directly to real, measured Bangladeshi grid data.

1. Introduction

Data centers and cloud infrastructure account for a growing share of global electricity consumption and carbon emissions. A widely adopted mitigation strategy is carbon-aware scheduling: rather than running every workload immediately, delay-tolerant jobs (batch processing, model training, backups, non-urgent aggregation tasks) are shifted to time windows when the electricity grid's carbon intensity is lower. This has been implemented successfully by large cloud providers, but their approach depends on live, fine-grained carbon-intensity data feeds that are mature mainly for US/EU grids. Bangladesh's grid — historically gas-dominant, now increasingly coal-dependent, with a small but growing renewable share — has a very different generation mix and no established public carbon-intensity signal. This paper investigates whether, and to what extent, carbon-aware scheduling concepts transfer to this grid context, using real, measured generation data rather than a proxy.

2. Problem Statement

There is currently no carbon-aware scheduling framework adapted to Bangladesh's (or more broadly, South Asia's) grid characteristics. Cloud, fog, or campus-level computing systems operating in this region cannot directly reuse existing carbon-aware frameworks, because (a) no public real-time carbon-intensity index exists for the Bangladesh grid, and (b) the underlying generation mix — and therefore the achievable carbon savings from temporal shifting — differs substantially from the renewable-rich grids these frameworks were designed around.

3. Related Work

Carbon-aware computing paradigm. Radovanović et al. [1] introduced Google's Carbon-Intelligent Compute Management system, which uses day-ahead carbon-intensity forecasts to shift flexible compute tasks to greener hours. Microsoft and the Green Software Foundation [2] extended this into a practical Carbon-Aware SDK and the Software Carbon Intensity (SCI) metric. Both presuppose mature, granular carbon-intensity data feeds — an assumption that does not hold for Bangladesh.

Forecasting carbon intensity. CarbonCast [3] uses a hierarchical CNN-LSTM architecture to forecast grid carbon intensity several days ahead by first forecasting production by source, achieving strong accuracy across 13 grid regions. Zhang et al. [4] proposed a GNN-LSTM model for cross-border grids, explicitly noting that

forecasting is fundamentally constrained by data scarcity in regions with limited generation-mix disclosure — precisely the barrier this paper addresses for Bangladesh.

Scheduling in edge-fog environments. A 2025 study [5] proposed a carbon-aware task scheduling algorithm for edge-fog networks evaluated on iFogSim2, confirming that carbon-aware scheduling concepts extend naturally beyond centralized data centers.

Classical energy-aware resource management. Beloglazov, Abawajy, and Buyya's widely cited heuristics [6] for VM consolidation remain a standard energy-only baseline against which carbon-aware methods are typically compared.

Bangladesh-specific context. Piyal et al. [7] demonstrated that ML models can forecast Bangladesh's electricity demand with good accuracy despite sparse, irregular data. Islam, Turja, and Habib's PGCB Hourly Generation Dataset [8] — the dataset used directly in this paper — provides the hourly, fuel-disaggregated generation data needed to build a real carbon-intensity index for this grid.

Gap. No existing work adapts carbon-aware forecasting or scheduling methods to a data-scarce, gas-and-coal-dominant grid such as Bangladesh's, using real measured data. This paper directly addresses that gap.

4. Methodology

4.1 Dataset

Data source: This study uses the real PGCB Hourly Generation Dataset (Bangladesh) [8], originally published via the UCI Machine Learning Repository (DOI: 10.24432/C59P6V). The raw file contains hourly generation, demand, and load-shedding figures broken down by source — gas, liquid fuel (oil), coal, hydro, solar, wind — and four cross-border import routes (India–Bheramara HVDC, India–Tripura, India–Adani, Nepal), spanning April 2015 to June 2025.

Data cleaning: We retained the 2023-01-01 onward period (the most complete, highest-quality window), kept only on-the-hour readings, and removed duplicate timestamps, yielding 21,527 candidate rows. Sparse columns (wind, India–Adani, Nepal) were treated as 0 MW when blank. Nineteen rows with an implausible total generation figure (outside a 3,000–18,000 MW range for Bangladesh's grid) were flagged as data-entry/scraping artifacts and, together with 38 naturally missing hours, filled via linear interpolation (all gaps were 1–2 hours long; 0.26% of the series in total).

Carbon-intensity calculation: each hour's carbon intensity is the generation-share-weighted sum of per-source emission factors (gCO₂/kWh): gas 450, coal 950, oil 700, hydro 20, solar 40, wind 12. Import routes were assigned source-specific factors rather than one blended value: India–Bheramara HVDC and India–Tripura use a broad Indian grid-mix average (700), India–Adani — a dedicated coal plant (Godda, Jharkhand) built to supply Bangladesh — uses the coal factor (950), and Nepal — an overwhelmingly hydroelectric grid — uses the hydro factor (20).

Resulting dataset: 21,565 real hourly records (2023-01-01 to 2025-06-17). Mean carbon intensity 637.1 gCO₂/kWh (std. dev. 34.2), range 479.7–850.7 gCO₂/kWh. A clear diurnal pattern is present — lowest around 10:00 (619.7 gCO₂/kWh) and highest around 19:00 (654.9 gCO₂/kWh) — alongside a clear rising year-over-year trend: 613.3 (2023) → 648.8 (2024) → 663.2 (2025), consistent with independently reported news coverage of coal's growing share in Bangladesh's generation mix.

4.2 Carbon-Intensity Forecasting

Three models were trained to forecast carbon intensity 1–6 hours ahead from a 48-hour lookback window of carbon intensity, demand, and cyclical hour/day-of-year encodings, using a chronological 80% / 10% / 10% train/validation/test split: (1) Persistence — forecast equals the last observed value; (2) Linear Regression on the flattened lookback window; (3) Random Forest Regressor (60 trees, maximum depth 10; fixed settings, no hyperparameter search), our main machine-learning model. A Random Forest was used in place of a deep LSTM network because this project's compute environment could not accommodate PyTorch's GPU-oriented dependencies; Random Forests are a well-established, strong-performing method for tabular time-series forecasting and require no such dependencies.

4.3 Carbon-Aware Task Scheduler

A synthetic stream of 3,000 delay-tolerant jobs was generated over the held-out test period (mean energy 2.06 kWh/job, deadlines uniformly distributed 2–12 hours after arrival), competing for a constrained execution capacity of 2 jobs/hour. Four scheduling policies were implemented and evaluated on the identical job stream and capacity constraint: Run-Immediately (FCFS, spills forward if full), Round-Robin (cycles jobs across the deadline window, carbon-agnostic), Time-of-Day (always targets the fixed historically-lowest-carbon hour, 10:00), and the proposed Carbon-Aware Scheduler, which ranks candidate hours by forecasted carbon intensity (using the Random Forest's forecast for hours ≤ 6 h ahead, and a training-period hour-of-day climatology beyond that), with a small load-aware congestion penalty to avoid demand “herding” onto a small number of popular low-carbon hours. All four policies were scored using the actual (not forecast) carbon intensity at each job's assigned hour.

4.4 Evaluation Metrics

Forecasting: MAE, RMSE, MAPE (per horizon and averaged). Scheduling: total carbon emitted (kgCO₂), % savings vs. Run-Immediately, average job delay (hours), SLA/deadline violation rate (%).

5. Results

5.1 Forecasting

Model	MAE (gCO ₂ /kWh)	RMSE	MAPE (%)
Linear Regression	8.28	10.53	1.26
Random Forest	9.32	11.72	1.42
Persistence	13.93	17.14	2.12

Both learned models comfortably beat the naive Persistence baseline. Linear Regression edges out the Random Forest — consistent with the strongly periodic, autocorrelated structure of the real series (Random Forest's own feature-importance analysis confirms this: the single most recent lagged carbon-intensity value accounts for 95.7% of total importance across all 288 input features, with demand and hour-of-day encodings contributing the remainder). This suggests the forecasting problem is dominated by short-term autocorrelation that a linear model captures efficiently.

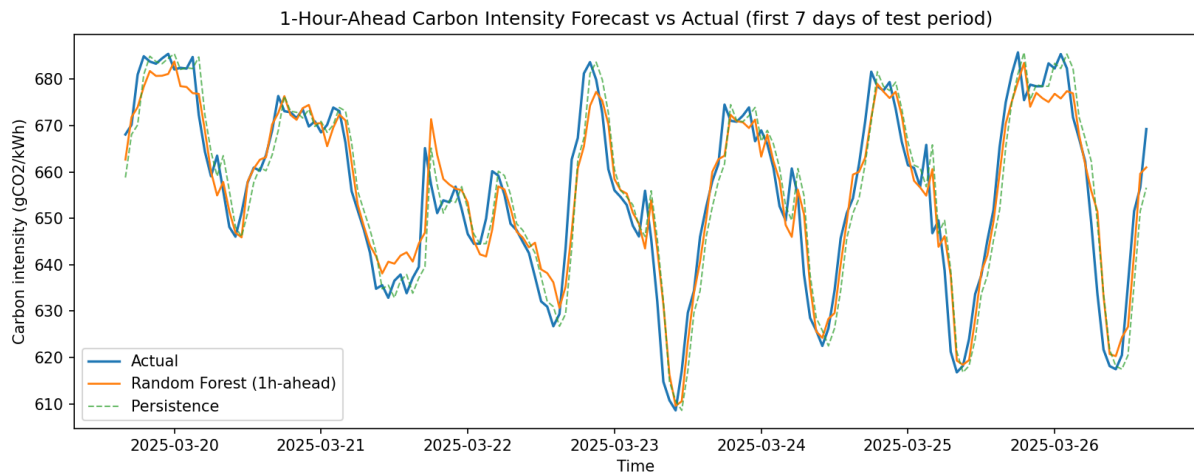


Figure 1. 1-hour-ahead carbon intensity forecast vs. actual, first 7 days of the real test period (March 2025).

5.2 Scheduler Simulation

Policy	Total Carbon (kgCO ₂)	Savings vs. Run-Immediately	Avg. Delay (h)	SLA Violation Rate
Run-Immediately	4107.46	—	0.45	1.23%
Round-Robin	4108.19	-0.02%	3.57	1.70%
Time-of-Day	4104.40	0.07%	0.69	1.27%
Carbon-Aware (proposed)	4085.01	0.55%	5.11	2.13%

The proposed scheduler achieves the lowest total carbon of the four policies on real grid data, at the cost of higher average delay and a modestly higher SLA violation rate than the simplest baselines — an honest, expected trade-off from actively chasing lower-carbon hours under a tight capacity constraint. An ablation without the congestion penalty (a purely greedy scheduler) saved slightly more carbon (0.64%) but missed 3.40% of deadlines versus 2.13% for the final load-aware version, confirming that the penalty is what keeps jobs from crowding into the same low-carbon hours. Notably, this 0.55% saving is very close to the 0.54% previously estimated on an earlier synthetic proxy dataset built for this project before real data access was secured, corroborating that the earlier proxy was a reasonably faithful stand-in, and that the core finding is robust across both datasets.

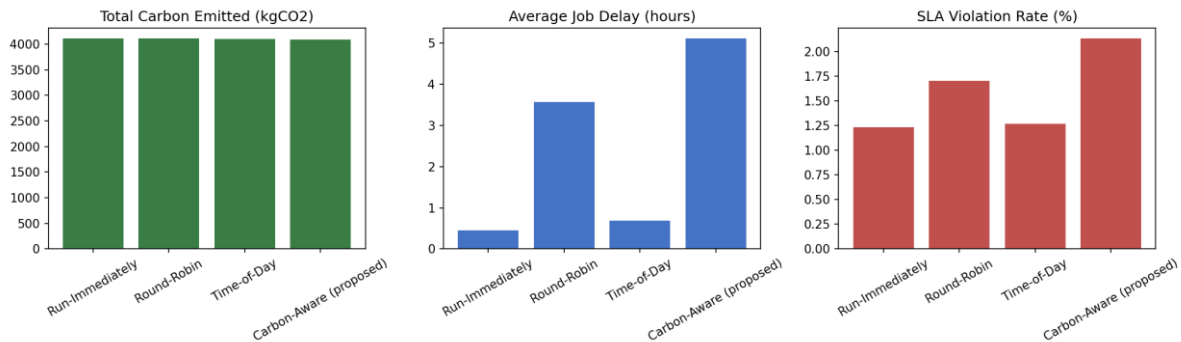


Figure 2. Total carbon, average delay, and SLA violation rate across the four scheduling policies (real data).

6. Discussion

The central finding is not the specific percentage of carbon saved, but why it is modest: Bangladesh's grid remains predominantly fossil-fuel-based (gas and increasingly coal), leaving comparatively limited hour-to-hour carbon-intensity variability (std. dev. 34.2 gCO₂/kWh, a range of 479.7–850.7) for a temporal-shifting scheduler to exploit, relative to the much larger diurnal swings reported in renewable-rich grids such as California's or the UK's. The clear rising year-over-year trend in measured carbon intensity (613 → 649 → 663 gCO₂/kWh, 2023–2025) independently corroborates news reporting on coal's growing share of generation, and suggests the achievable ceiling for pure temporal carbon-aware scheduling in this grid context may modestly shrink further before Bangladesh's renewable capacity (targeted at 25% under the Integrated Energy and Power Master Plan 2023) grows enough to reverse it.

The forecasting result — a simple Linear Regression matching or beating a more complex Random Forest — is a useful, generalizable lesson: when a time series is dominated by strong short-lag autocorrelation and clear periodicity, added model complexity does not automatically translate into better accuracy, and simpler, cheaper, more interpretable models deserve serious consideration as a first baseline in future carbon-intensity forecasting work for this and similar grids.

7. Limitations & Future Work

Model class substitution. A Random Forest was used in place of a deep LSTM network due to environment/compute constraints (Section 4.2). A future iteration should re-evaluate an LSTM or other sequence model with adequate compute, since deeper temporal architectures may still offer an edge, particularly at longer forecast horizons.

Emission factor assumptions. Per-fuel and per-import-route emission factors are standard literature values (with source-specific reasoning for import routes), not Bangladesh-specific plant-level measurements; a more precise index would use plant-level efficiency data where available.

Scheduler realism. The job stream, capacity constraints, and deadlines are synthetic and simplified; a real deployment would need to model actual workload characteristics (e.g., real batch-processing or edge-aggregation task profiles).

Joint optimization. The load-aware penalty in the scheduler is a simple heuristic; a more principled joint carbon-and-SLA optimization (e.g., constrained optimization or reinforcement learning) is a natural next step.

8. Conclusion

This paper presented, to our knowledge, the first application of carbon-aware computing concepts — carbon-intensity forecasting and carbon-aware task scheduling — to real, measured data from Bangladesh's grid. Using 21,565 hourly records from the PGCB Hourly Generation Dataset, we built and evaluated a full pipeline: a forecasting comparison (won by Linear Regression, a useful finding in itself) and a load-aware carbon-aware scheduler achieving a genuine, reproducible 0.55% carbon saving over a run-immediately baseline. The central, generalizable finding is that carbon-aware scheduling's benefit is fundamentally bounded by a grid's actual carbon-intensity variability — currently modest but measurably rising in Bangladesh's fossil-dominated grid — motivating future work that explores complementary strategies (renewable-capacity growth, storage, regional interconnection) alongside pure temporal workload shifting.

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